

PLC market segments

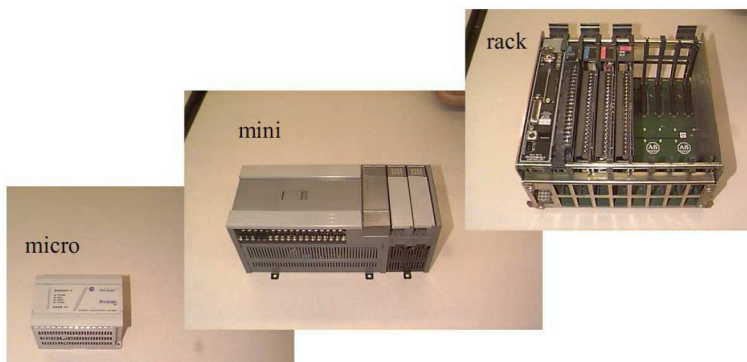
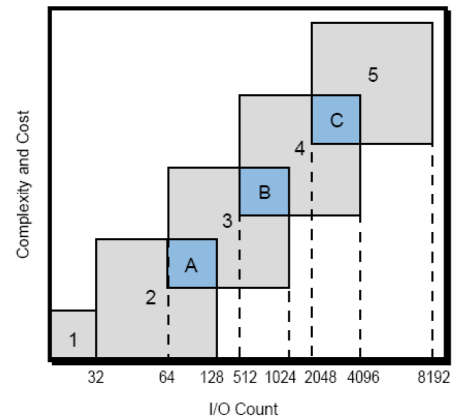
Controller classes

- Microcontrollers
- Small controllers
- Medium drivers
- Large controllers
- Very large controllers

Siemens controllers

- Logo!
- Family S7-1200, S7-200, S7-200
- Family S7-1500, S7-300
- Family S7-400

PLC market segments



Driver market segments



Allen Controller Family -
Bradley

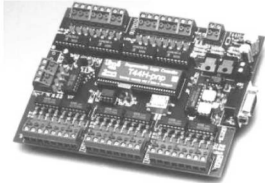


The smallest Mitsubishi driver

Construction types of controllers

- Compact design - complete system with limited expansion possibilities
- Modular design - flexibly configurable combination of central unit and input/output modules
- Controllers integrated with the operator panel
- Indirect - advantages of compact and modular (SAIA)
- Without its own housing - open frame

Construction types of controllers



Without housing -
Triangle Research



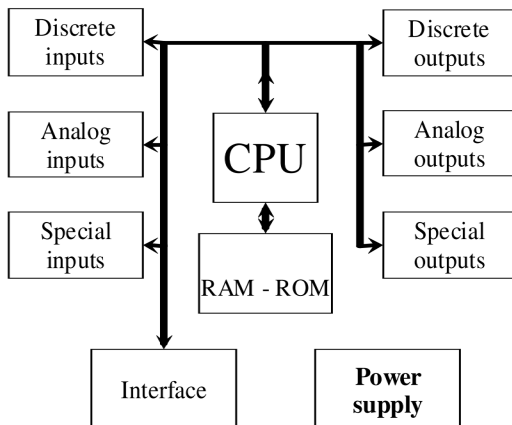
Compact design -
Mitsubishi FX0



Modular construction - Simatic S7 - 300



Controller structure



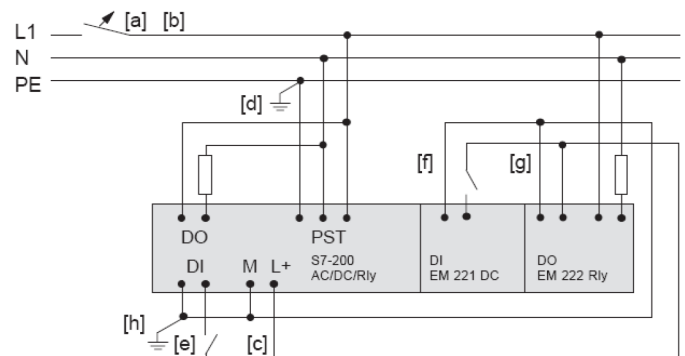
Controller processor

- CPU - *Central Processor Unit* - microprocessor and environmental chipsets
- The processors used range from 8-bit straight (Modicon Micro) to 32-bit complex (Modicon TSX Quantum).
- Sometimes - multiprocessor chips with specialized processors.

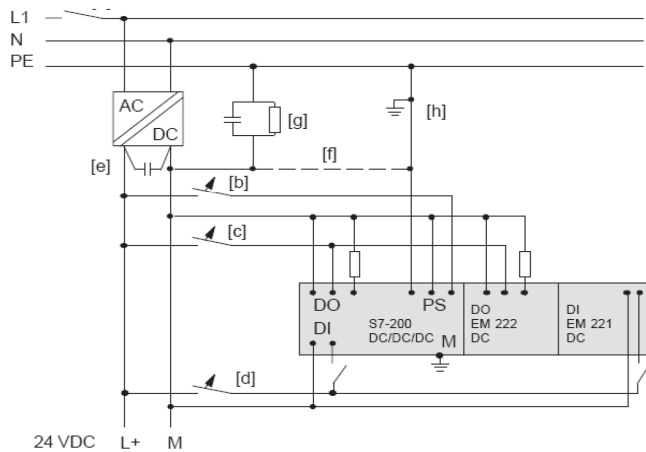
Controller memory

- ROM (Read Only Memory) type memory - a memory for writing only, stores the content permanently. Currently, the most common types of EPROM or EEPROM are
- RAM (Random Access Memory) type memory for writing and reading. Loses content when power is turned off; can be battery backed up.
- Flash memory - EEPROM for fast erase and write

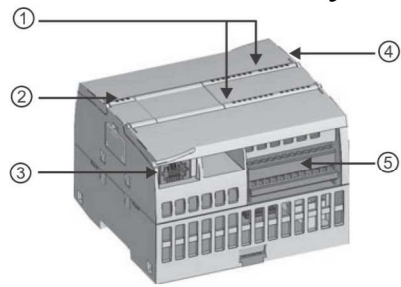
Built-in power supply - compact design



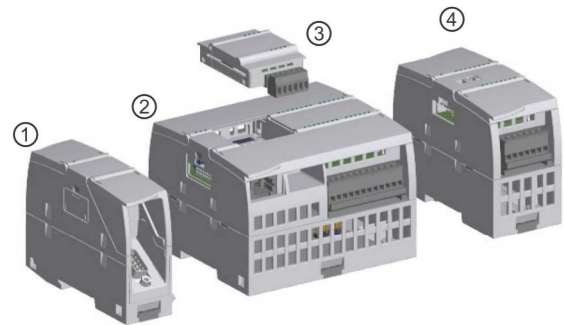
External power supply



Hardware configuration of S7-1200 family of controllers



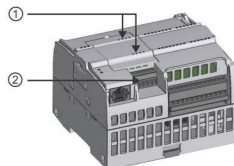
Feature	CPU 1211C	CPU 1212C	CPU 1214C	CPU 1215C
Physical size (mm)	90 x 100 x 75	90 x 100 x 75	110 x 100 x 75	130 x 100 x 75
User memory				
Work	30 Kbytes	50 Kbytes	75 Kbytes	100 Kbytes
Load	1 Mbyte	1 Mbyte	4 Mbytes	4 Mbytes
Retentive	10 Kbytes	10 Kbytes	10 Kbytes	10 Kbytes
Local on-board I/O	Digital	6 inputs/4 outputs	8 inputs/6 outputs	14 inputs/10 outputs
	Analog	2 inputs	2 inputs	2 inputs / 2 outputs
Process image size	Inputs (I)	1024 bytes	1024 bytes	1024 bytes
	Outputs (Q)	1024 bytes	1024 bytes	1024 bytes
Bit memory (M)	4096 bytes	4096 bytes	8192 bytes	8192 bytes
Signal module (SM) expansion	None	2	8	8
Signal board (SB), Battery board (BB), or communication board (CB)	1	1	1	1
Communication module (CM) (left-side expansion)	3	3	3	3
High-speed counters	Total	3 built-in I/O, 5 with SB	4 built-in I/O, 6 with SB	6
	Single phase	3 at 100 kHz SB: 2 at 30 kHz	3 at 100 kHz 1 at 30 kHz SB: 2 at 30 kHz	3 at 100 kHz 3 at 30 kHz
	Quadrature phase	3 at 80 kHz SB: 2 at 20 kHz	3 at 80 kHz 1 at 20 kHz SB: 2 at 20 kHz	3 at 80 kHz 3 at 20 kHz
Pulse outputs ¹	4	4	4	4
Memory card	SIMATIC Memory card (optional)			
Real time clock retention time	20 days, typ. / 12 day min. at 40 degrees C (maintenance-free Super Capacitor)			
PROFINET	1 Ethernet communication port			2 Ethernet communication ports
Real math execution speed	2.3 µs/instruction			
Boolean execution speed	0.08 µs/instruction			



- ① Communication module (CM), communication processor (CP), or TS Adapter
- ② CPU
- ③ Signal board (SB), communication board (CB), or Battery Board (BB)
- ④ Signal module (SM)

The CPU supports one plug-in expansion board:

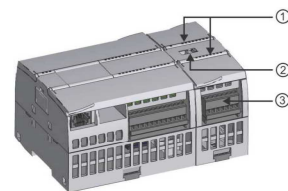
- A signal board (SB) provides additional I/O for your CPU. The SB connects on the front of the CPU.
- A communication board (CB) allows you to add another communication port to your CPU.
- A battery board (BB) allows you to provide long term backup of the realtime clock.



- ① Status LEDs on the SB
- ② Removable user wiring connector

Signal modules (SMs) add additional functionality to the CPU. SMs connect to the right side of the CPU.

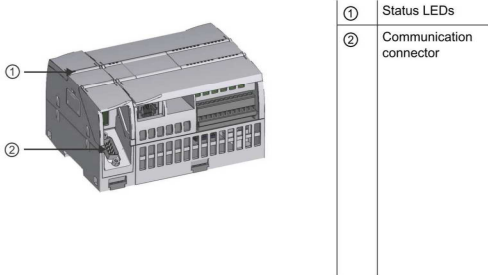
- Digital I/O
- Analog I/O
- RTD and thermocouple



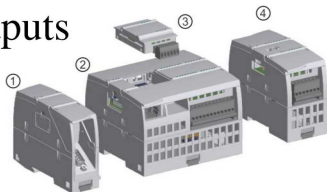
- ① Status LEDs
- ② Bus connector
- ③ Removable user wiring connector

Communication modules (CMs) and communications processors (CPs) add communication options to the CPU, such as for PROFIBUS or RS232 / RS485 connectivity (for PIP, Modbus or USS), or the AS-i master. A CP provides capabilities for other types of communication, such as to connect the CPU over a GPRS network.

- The CPU supports up to 3 CMs or CPs
- Each CM or CP connects to the left side of the CPU (or to the left side of another CM or CP)

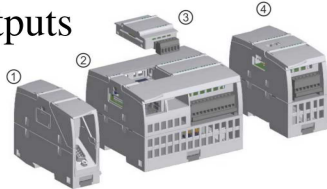


Controller extensions - digital inputs and outputs



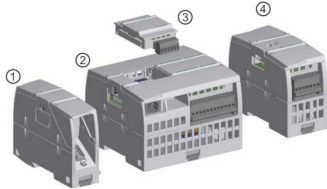
Type	Input only	Output only	Combination In/Out
③ digital SB	• 4 x 24VDC In, 200 kHz • 4 x 5VDC In, 200 kHz	• 4 x 24VDC Out, 200 kHz • 4 x 5VDC Out, 200 kHz	• 2 x 24VDC In / 2 x 24VDC Out • 2 x 24VDC In / 2 x 24VDC Out, 200 kHz • 2 x 5VDC In / 2 x 5VDC Out, 200 kHz
④ digital SM	• 8 x 24VDC In • 16 x 24VDC In	• 8 x 24VDC Out • 8 x Relay Out • 8 x Relay Out (Changeover) • 16 x 24VDC Out • 16 x Relay Out	• 8 x 24VDC In / 8 x 24VDC Out • 8 x 24VDC In / 8 x Relay Out • 8 x 120/230VAC In / 8 x Relay Out • 16 x 24VDC In / 16 x 24VDC Out • 16 x 24VDC In / 16 x Relay Out

Controller extensions - analog inputs and outputs



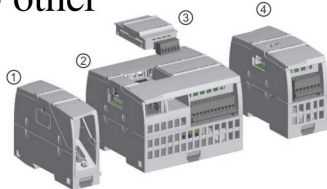
Type	Input only	Output only	Combination In/Out
③ analog SB	• 1 x 12 bit Analog In • 1 x 16 bit RTD • 1 x 16 bit Thermocouple	• 1 x Analog Out	-
④ analog SM	• 4 x Analog In • 4 x Analog In x 16 bit • 8 x Analog In • Thermocouple: – 4 x 16 bit TC – 8 x 16 bit TC • RTD: – 4 x 16 bit RTD – 8 x 16 bit RTD	• 2 x Analog Out • 4 x Analog Out	• 4 x Analog In / 2 x Analog Out

Controller extensions - communication



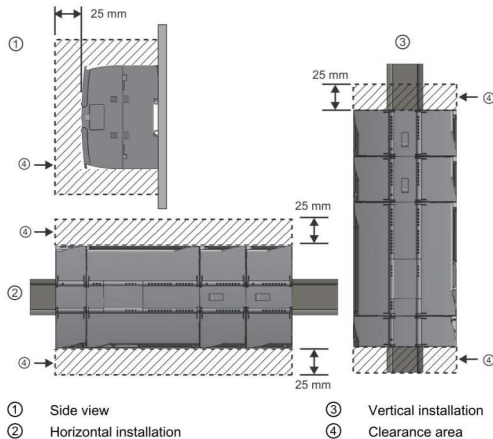
Module	Type	Description
① Communication module (CM)	RS232	Full-duplex
	RS422/485	Full-duplex (RS422) Half-duplex (RS485)
	PROFIBUS Master	DPV1
	PROFIBUS Slave	DPV1
	AS-i Master (CM 1243-2)	AS-Interface
① Communication processor (CP)	Modem connectivity	GPRS
① Communication board (CB)	RS485	Half-duplex
① TeleService	TS Adapter IIE Basic ¹	Connection to CPU
	TS Adapter GSM	GSM/GPRS
	TS Adapter Modem	Modem
	TS Adapter ISDN	ISDN
	TS Adapter RS232	RS232

Controller extensions - other



Module	Description
③ Battery board	Plugs into expansion board interface on front of CPU. Provides long term backup of realtime clock

Installation location



Working conditions

Environmental conditions - Operating	
Ambient temperature range (Inlet Air 25 mm below unit)	-20° C to 60° C horizontal mounting -20° C to 50° C vertical mounting 95% non-condensing humidity Unless otherwise specified
Atmospheric pressure	1080 to 795 hPa (corresponding to an altitude of -1000 to 2000m)
Concentration of contaminants	SO ₂ : < 0.5 ppm; H ₂ S: < 0.1 ppm; RH < 60% non-condensing
EN 60068-2-14, Test Nb, temperature change	5° C to 55° C, 3° C/minute
EN 60068-2-27 Mechanical shock	15 G, 11 ms pulse, 6 shocks in each of 3 axis
EN 60068-2-6 Sinusoidal vibration	DIN rail mount: 3.5 mm from 5-9 Hz, 1G from 9 - 150 Hz Panel Mount: 7.0 mm from 5-9 Hz, 2G from 9 to 150 Hz 10 sweeps each axis, 1 octave per minute

NOTICE

For systems that must start up in the range of -20° C to 0° C, the user program should delay energizing outputs for 10 seconds following startup.

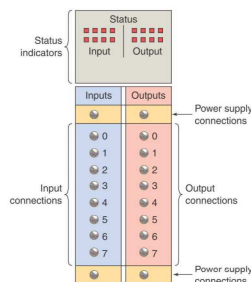


Figure 2-11 16, 32, and 64 point I/O modules.
Source: (a1) Photos courtesy Omron Industrial Automation, www.ia.omron.com

Parameters of the input modules

- For DC inputs - polarity:
 - positive logic** (sink in, PNP) - switched pole (+), common terminal (-)
 - negative logic** (source in, NPN) switched pole (-), common terminal (+)
- Opto-insulation, r.m.s. value of insulation breakdown voltage, time of guaranteed protection
- Hardware and software signal filtering - elimination of short pulses

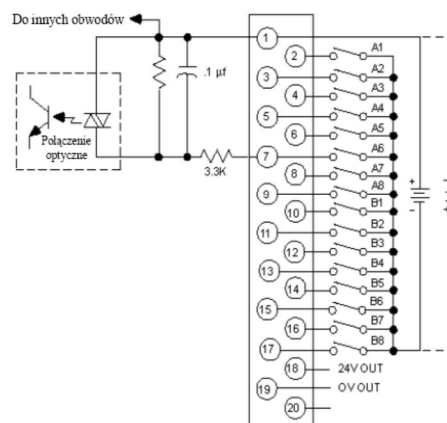
- Configuration - number of electrically independent groups, size of groups
- ON Response* and *OFF Response* times
- Current consumption (important for power supply selection)
- Operating conditions - temperature, humidity, vibrations
- Electrical characteristics - in the ON and OFF state

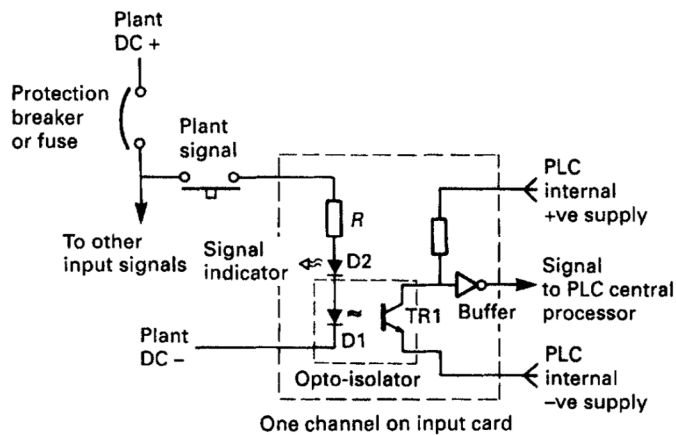
Discrete inputs of PLCs

- Type DC

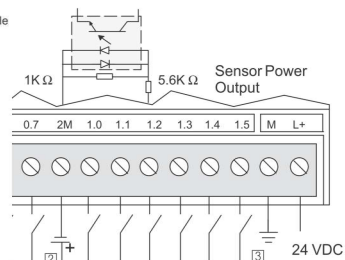
Input type	Sink/Source (IEC Type 1 sink)
Input Voltage	
Maximum continuous permissible	30 VDC
Surge	35 VDC for 0.5 s
Rated value	24 VDC at 4 mA, nominal
Logic 1 signal (minimum)	15 VDC at 2.5 mA, minimum
Logic 0 signal (maximum)	5 VDC at 1 mA, maximum
Isolation (Field Side to Logic Circuit)	
Optical isolation (galvanic)	500 VAC for 1 minute
Isolation groups of	4 points/2 points
Input Delay Times	
Filtered Inputs and Interrupt Inputs	0.2 to 12.8 ms, user-selectable

DC inputs





Description Order Number	CPU 224 DC/DC/DC 6ES7214-1AD20-0XB0
Input Features	
Number of integrated inputs	14 inputs
Input type	Sink/Source (IEC Type 1)
Input Voltage	
Maximum continuous permissible	30 VDC
Surge	35 VDC for 0.5 s
Rated value	24 VDC at 4 mA, nominal
Logic 1 signal (minimum)	15 VDC at 2.5 mA, minimum
Logic 0 signal (maximum)	5 VDC at 1 mA, maximum
Isolation (Field Side to Logic Circuit)	
Optical isolation (galvanic)	500 VAC for 1 minute
Isolation groups of	8 points and 6 points
Input Delay Times	
Filtered inputs and interrupt inputs	0.2 to 12.8 ms, user-selectable
HSC clock input rate	
Single Phase	
Logic 1 level = 15 to 30 VDC	20 kHz
Logic 1 level = 15 to 26 VDC	30 kHz
Quadrature	
Logic 1 level = 15 to 30 VDC	10 kHz
Logic 1 level = 15 to 26 VDC	20 kHz
Connection of 2 Wire Proximity Sensor (Bero)	
Permissible leakage current	1 mA, maximum
Cable Length	
Unshielded (not HSC)	300 m
Shielded	500 m
HSC inputs, shielded	50 m

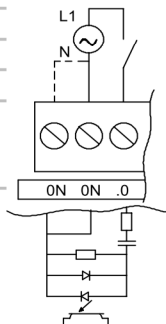


Discrete inputs of PLCs

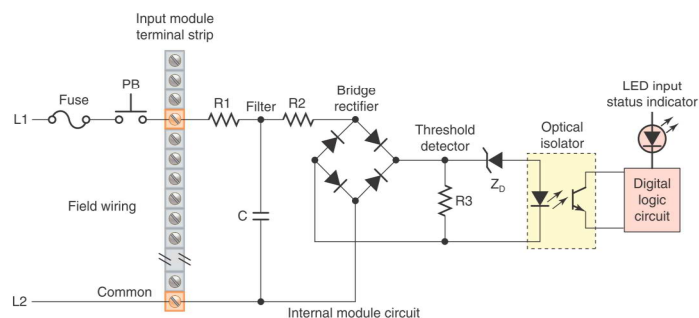
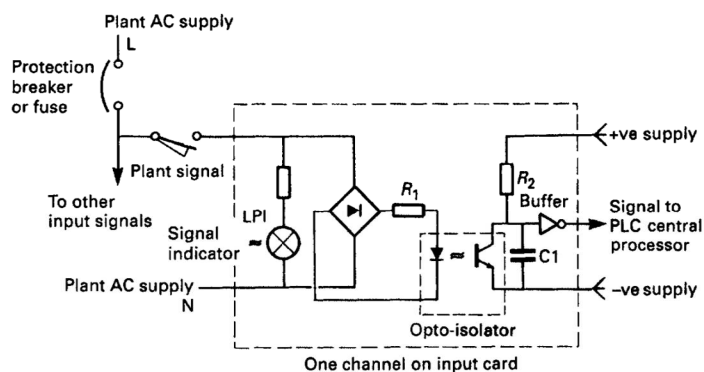
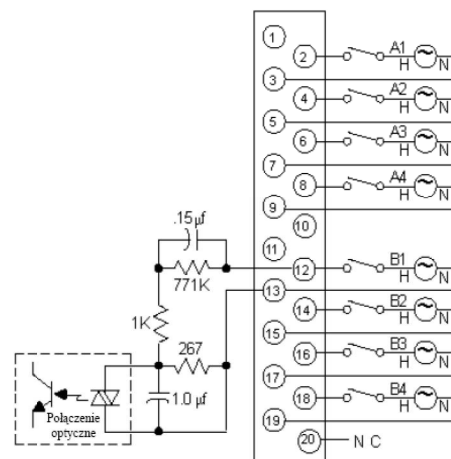
• Type AC

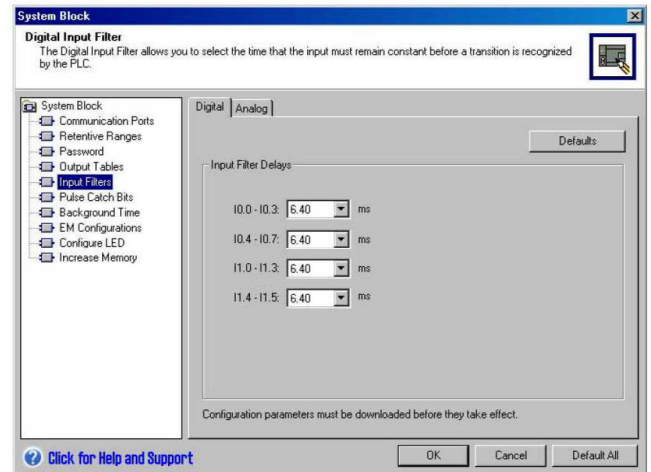
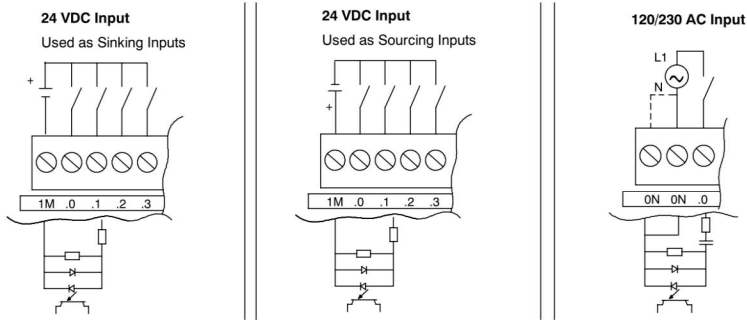
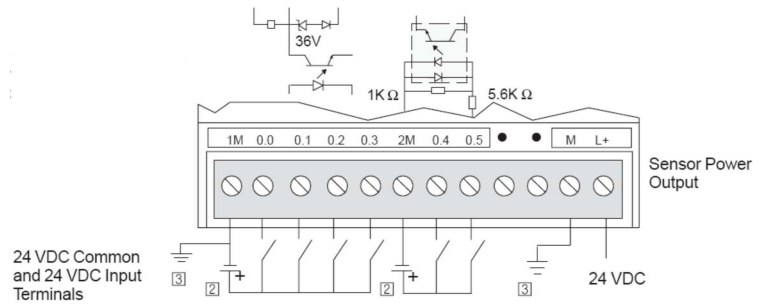
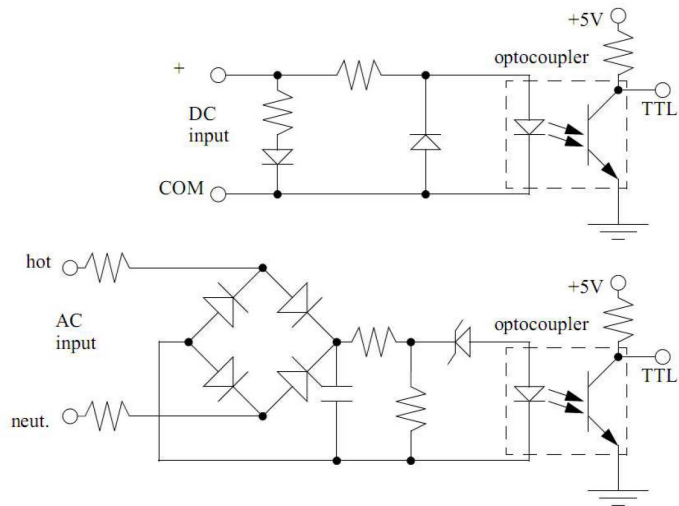
Rated voltage	120 VAC at 6 mA or 230 VAC at 9 mA nominal
Maximum continuous permissible voltage	264 VAC
Surge voltage (max.)	-
Logic 1 (min.)	79 VAC at 2.5 mA
Logic 0 (max.)	20 VAC or 1 mA AC
Input delay (max.)	15 ms
Connection of 2 wire proximity sensor (Bero)	
Permissible leakage current (max.)	1 mA AC
Isolation	
Optical (galvanic, field to logic)	1500 VAC for 1 minute
Isolation groups	1 point

120/230 AC Input



AC inputs





0.2 - 12.8 ms

Short pulse detection

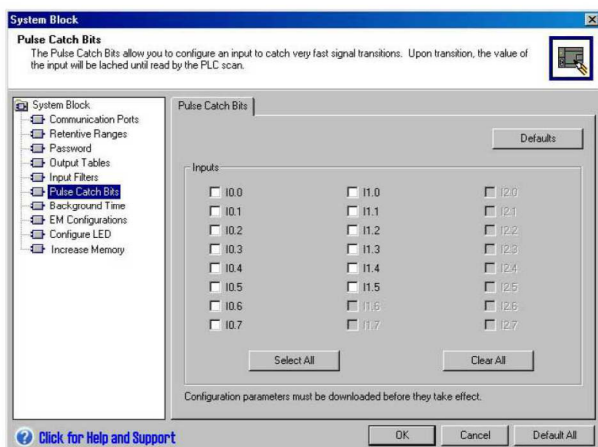


Figure 4-25 Pulse Catch

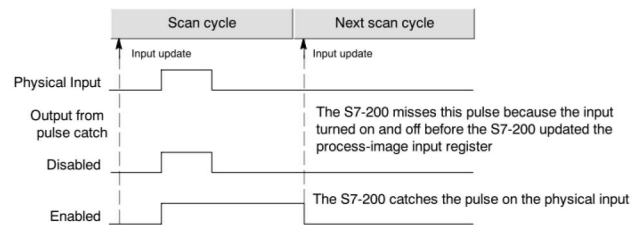


Figure 4-26 Operation of the S7-200 with the Pulse Catch Feature Enabled and Disabled

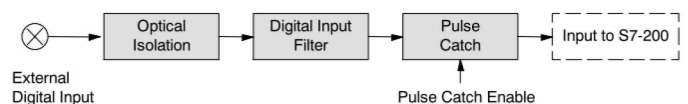


Figure 4-27 Digital Input Circuit

Sensor types connected to PLC inputs

- With relay output
- With transistor output
- With alternating current output
- Other - conversion required

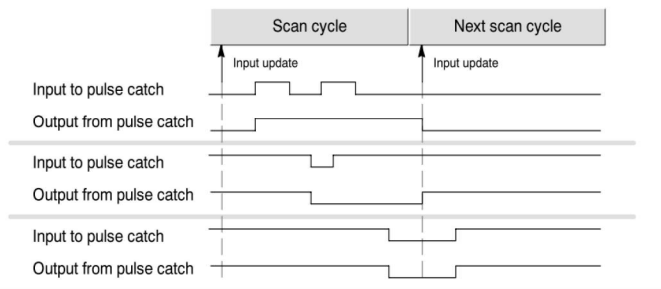
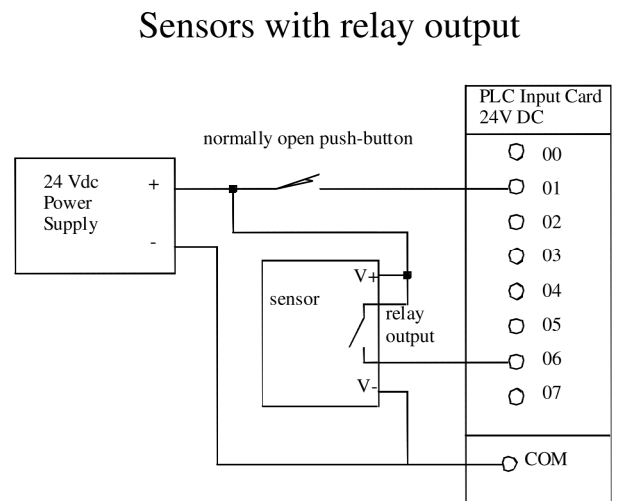
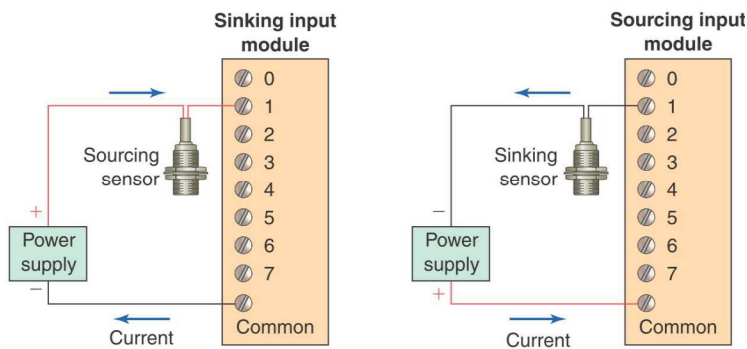
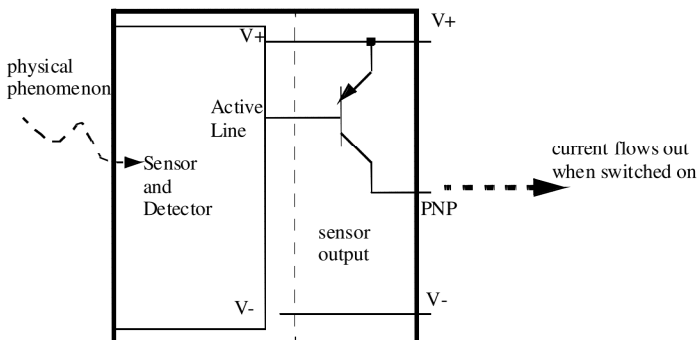


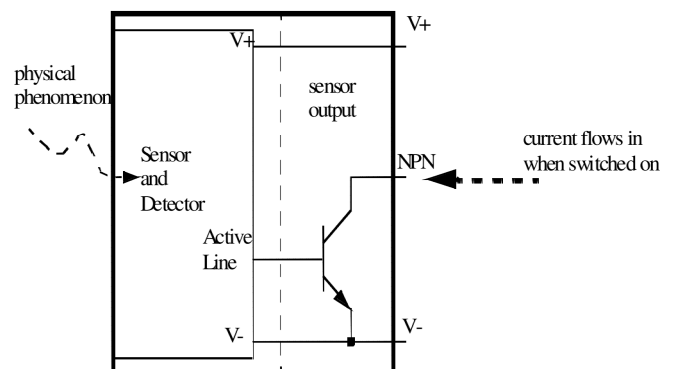
Figure 4-28 Responses of the Pulse Catch Function to Various Input Conditions



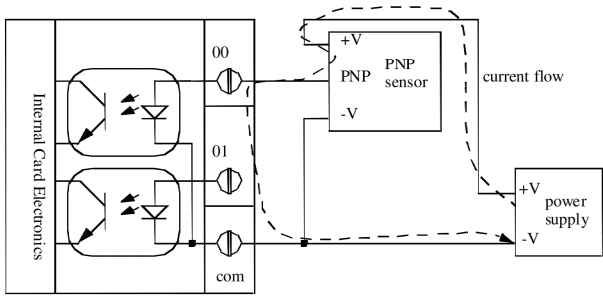
Type of 'source' -sourcing



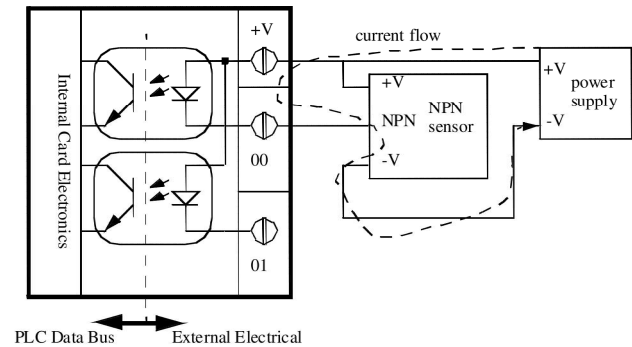
The "drain" type. - sinking



Connection to PLC - PNP sensor (source)



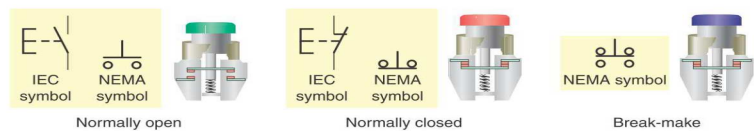
Connection to PLC - NPN sensor (drain)



Sensor	Advantages	Disadvantages	Applications
Limit Switch	- High Current Capability - Low Cost - Familiar "Low-Tech" Sensing	- Requires Physical Contact with Target - Very Slow Response - Contact Bounce	- Interlocking - Basic End-of-Travel Sensing
Photoelectric	- Senses all Kinds of Materials - Long Life - Longest Sensing Range - Very Fast Response Time	- Lens Subject to Contamination - Sensing Range Affected by Color and Reflectivity of Target	- Packaging - Material Handling - Parts Detection

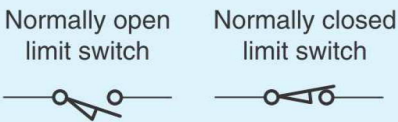
Sensor	Advantages	Disadvantages	Applications
Inductive	- Resistant to Harsh Environments - Very Predictable - Long Life - Easy to Install	Distance Limitations	- Industrial and Machines - Machine Tool - Senses Metal-Only Targets
Capacitive	- Detects Through Some Containers - Can Detect Non-Metallic Targets	Very Sensitive to Extreme Environmental Changes	Level Sensing
Ultrasonic	Senses all Materials	- Resolution - Repeatability - Sensitive to Temperature Changes	- Anti-Collision - Doors - Web Brake - Level Control

Buttons

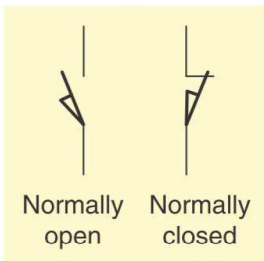


Contacts

NEMA symbols

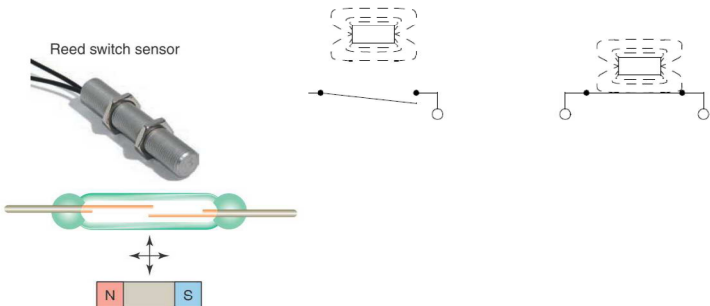


IEC symbols

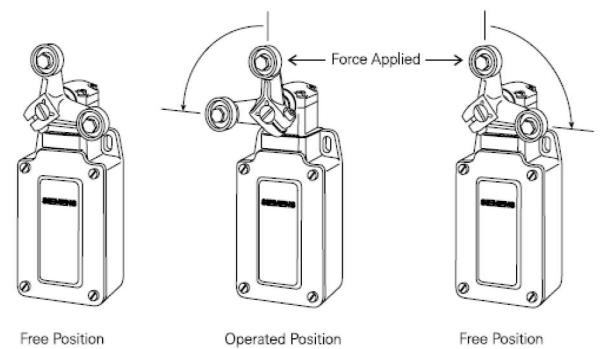
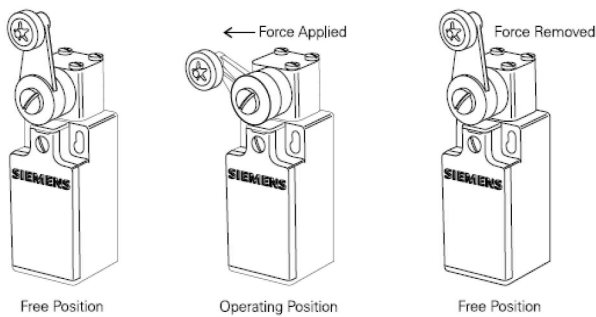
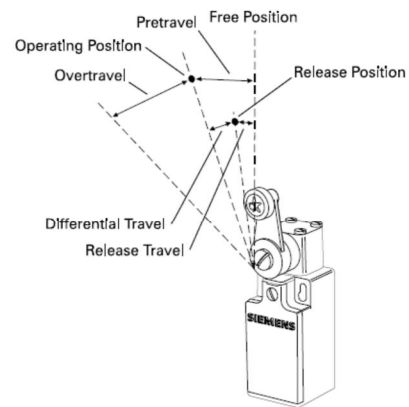
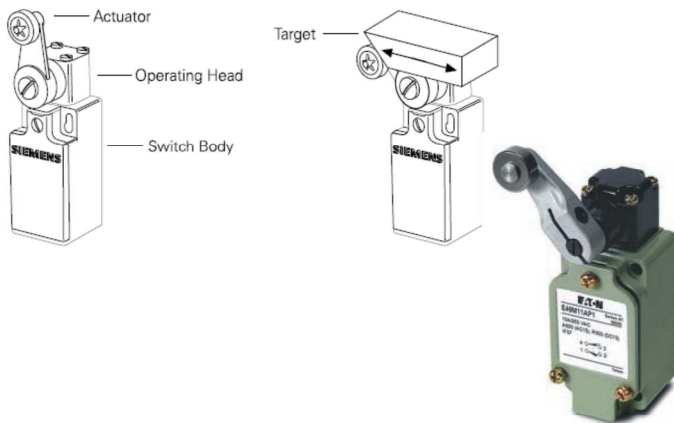


Detection of objects

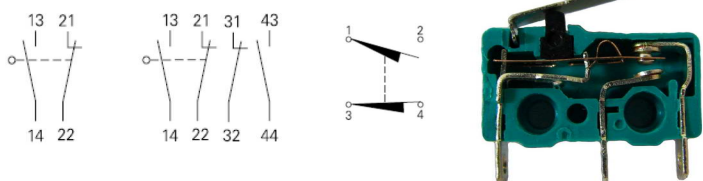
- Mechanical contact sensors
 - The force between the sensor and the object is displayed.
 - Limit switches - overrunning, collision,...
- Contracts



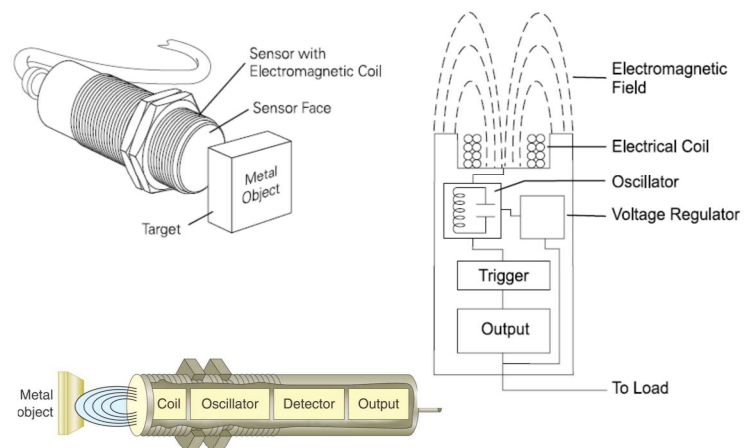
Limit switches



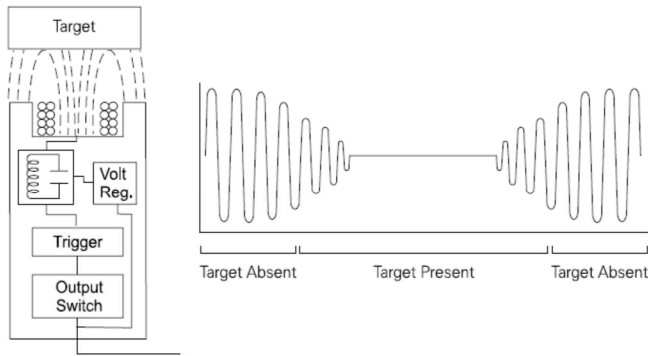
Contact State	Break-Before-Make		Make-Before-Break	
	NO	NC	NO	NC
Free Position	Open	Closed	Open	Closed
Transition	Open	Open	Closed	Closed
Operated State	Closed	Open	Closed	Open



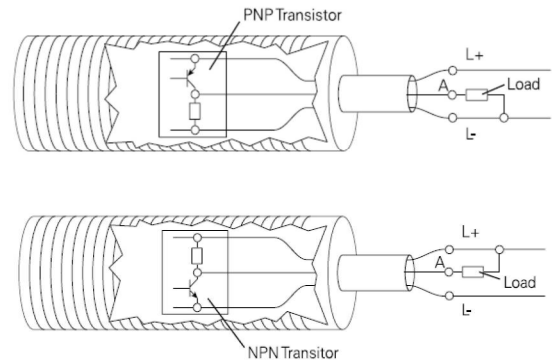
Proximity sensors: inductive



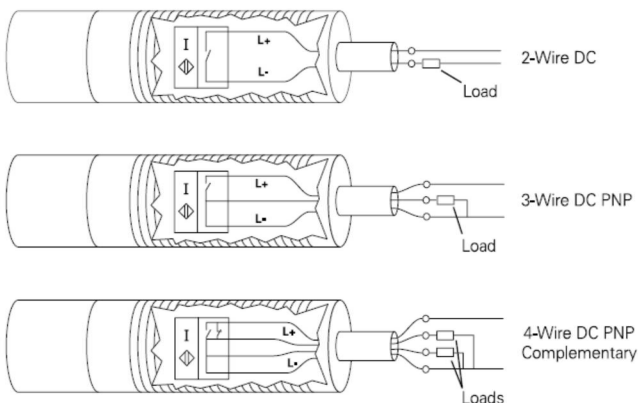
Inductive sensors - principle of operation



Connection method



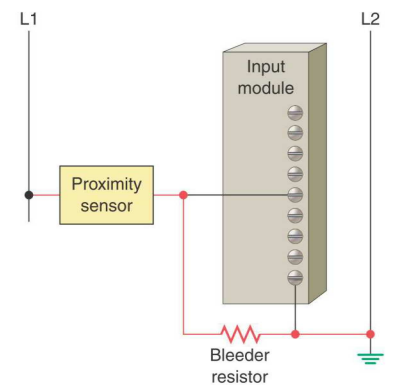
2-, 3- and 4-wire configuration



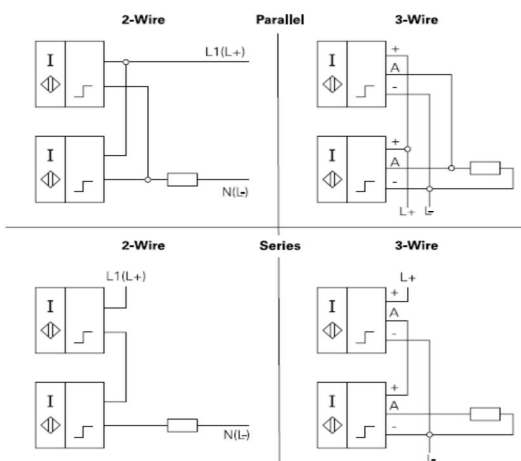
2-wire configuration

$$I_{\text{off}} \leq 1...2 \text{ mA}$$

$$I_{\text{on}} > 4 \text{ mA}$$

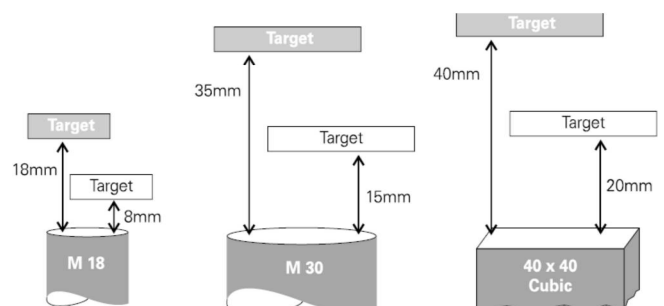


Connection of sensors

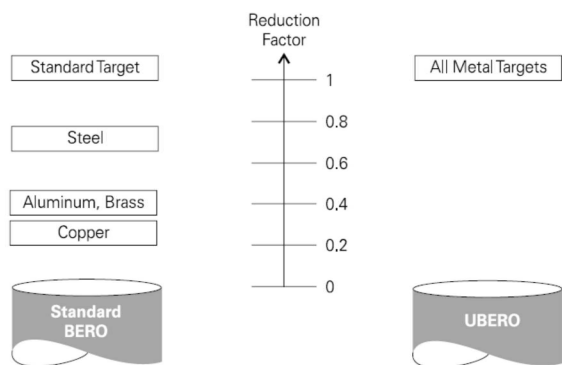


Nominal sensor activation distance

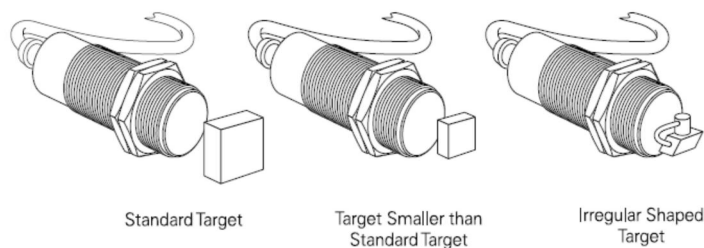
- Standard version - 0.6 to 20 mm
- Special version 1 to 65 mm



Response distance depends on the material

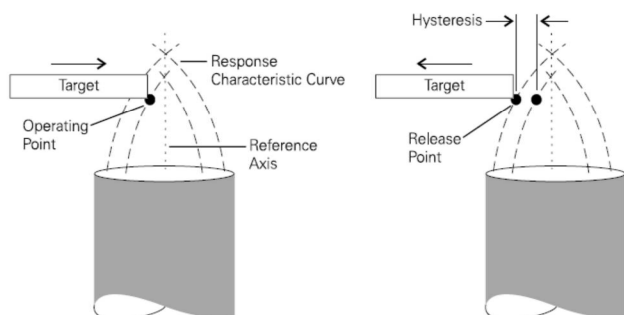


The area of action depends on the size of the object

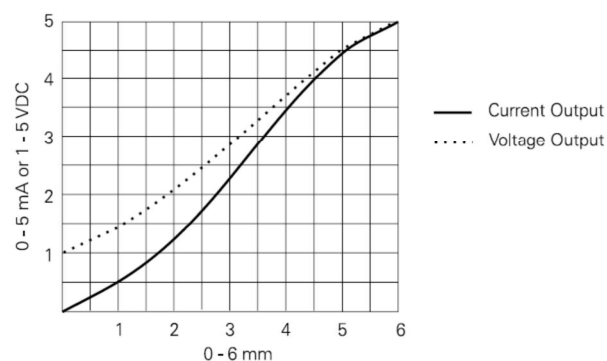


Size of Target Compared to Standard Target	Correction Factor	
	Shielded	Unshielded
25%	0.56	0.50
50%	0.83	0.73
75%	0.92	0.90
100%	1.00	1.00

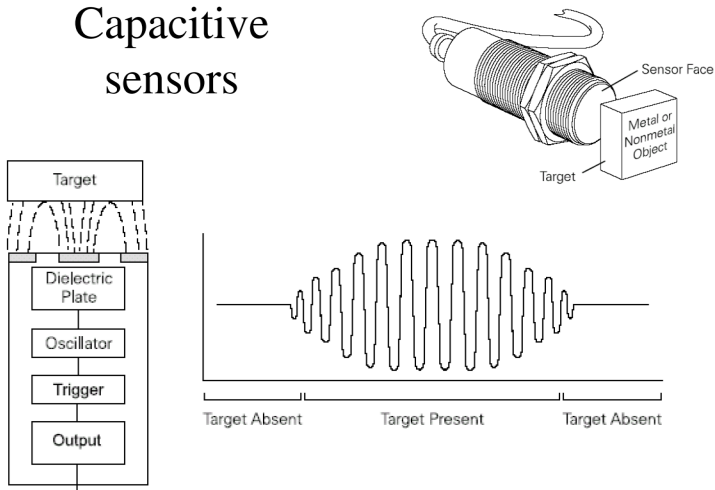
Detector activation hysteresis zone



Sensors with analog output



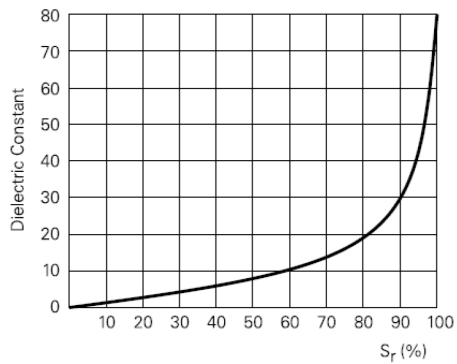
Capacitive sensors



Selection of sensors

Housing Dimension (mm)	Material	Shielded Unshielded	Sn (mm)	Operating Voltage	Wires
18	Plastic	Shielded	5	10-65 VDC	3
30	Metal	Shielded	10	20-250 VAC	3
	Plastic	Shielded	10	20-250 VAC	2
	Metal	Shielded	10	10-65 VDC	4
	Plastic	Shielded	10	10-65 VDC	4
40	Plastic	Shielded	20	20-250 VAC	2
	Plastic	Shielded	20	10-65 VDC	4
40x40 (Limit Switch Style)	Plastic	Shielded	20	20-250 VAC	2
	Plastic	Shielded	20	10-65 VDC	4
20x20 (Flat Pack)	Metal	Shielded	5	10-30 VDC	3

Area of activity



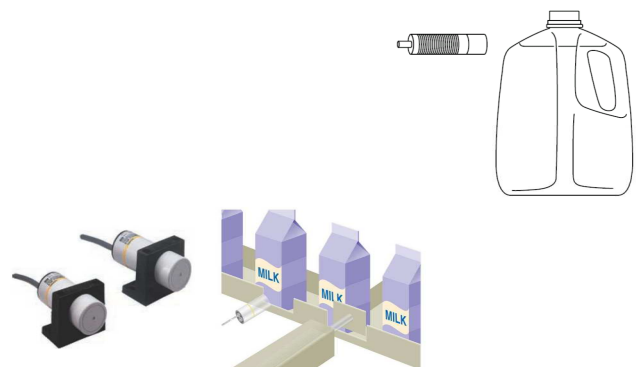
Permeability constant

Material	Dielectric Constant	Material	Dielectric Constant
Alcohol	25.8	Polyamide	5
Araldite	3.6	Polyethylene	2.3
Bakelite	3.6	Polypropylene	2.3
Glass	5	Polystyrene	3
Mica	6	Polyvinyl Chloride	2.9
Hard Rubber	4	Porcelain	4.4
Paper-Based Laminate	4.5	Pressboard	4
Wood	2.7	Silica Glass	3.7
Cable Casting Compound	2.5	Silica Sand	4.5
Air, Vacuum	1	Silicone Rubber	2.8
Marble	8	Teflon	2
Oil-Impregnated Paper	4	Turpentine Oil	2.2
Paper	2.3	Transformer Oil	2.2
Paraffin	2.2	Water	80
Petroleum	2.2	Soft Rubber	2.5
Plexiglas	3.2	Celluloid	3

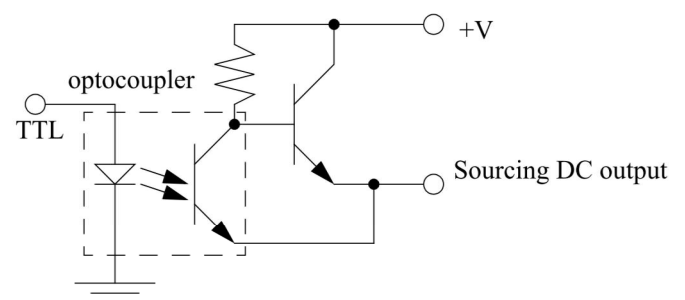
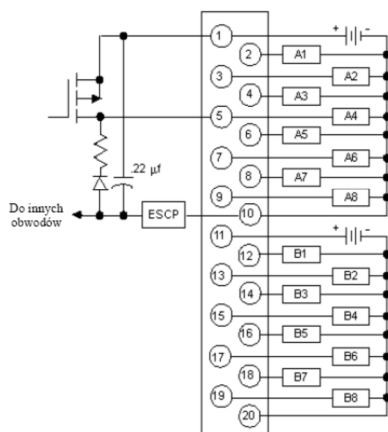
Substancja	ϵ_r
próżnia	1
powietrze	1,00054
teflon	2,1
polietylen	2,25
polistyren	2,4–2,7
dwusiarzek węgla	2,6
lód	100
papier	3,5
krzemionka	3,7
beton	4,5
guma	7
diamant	5,5–10
sól kuchenna	3–15
grafit	10–15
krzem	11,68

amoniak	26–22–20–17 (–80 ÷ –40 ÷ 0 ÷ 20 °C)
alkohol metylowy	30
gliceryna	41,2–47–42,5 (0–20–25 °C)
woda	88–80,1–55,3–34,5 (0–20–100–200 °C)
kwask fluorowodorowy	83,6 (0 °C)
kwask siarkowy	84–100 (20–25 °C)
cyjanowodór	158,0–2,3 (0–21 °C)
tlenek tytanu(IV)	86–173
ferroelektryki	>1000

Detection (e.g. water) through barriers



DC outputs



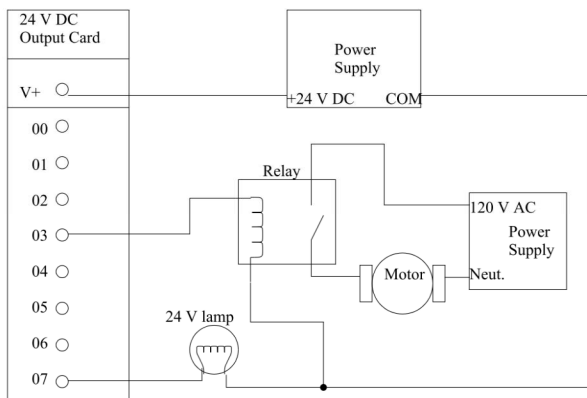
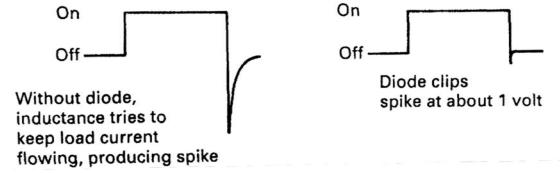
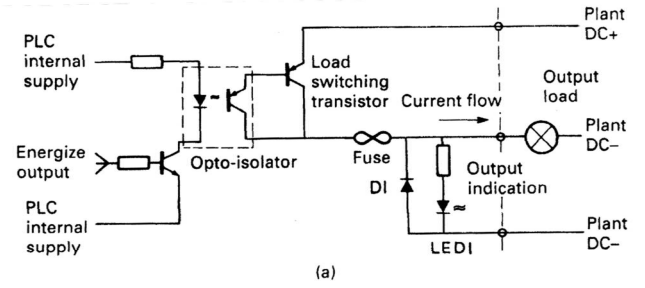
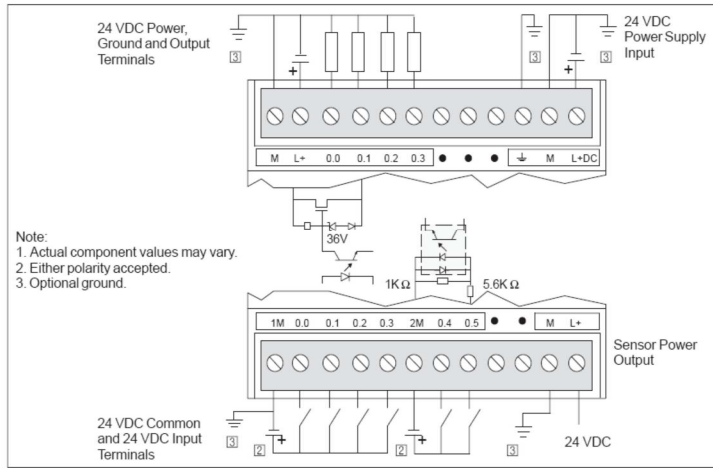
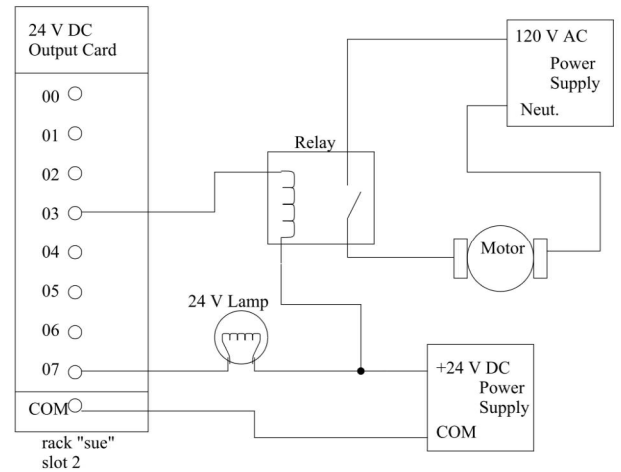
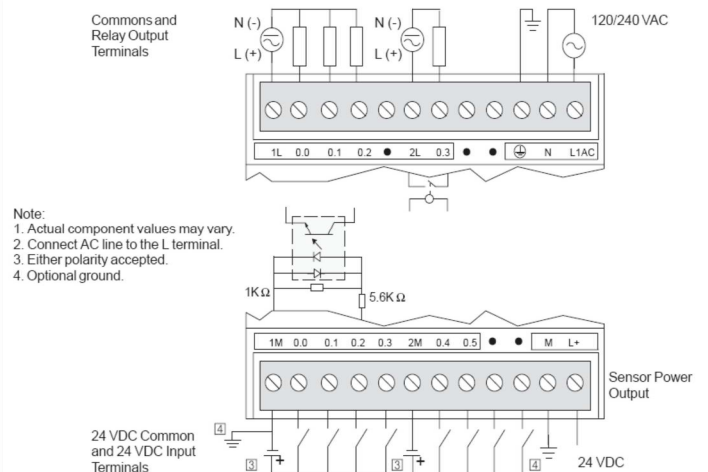
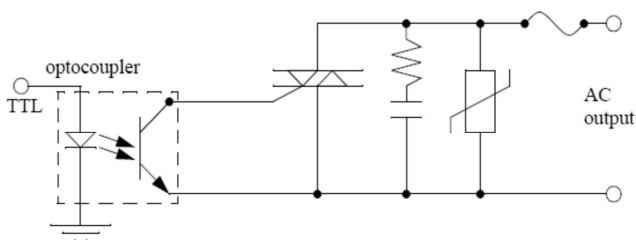


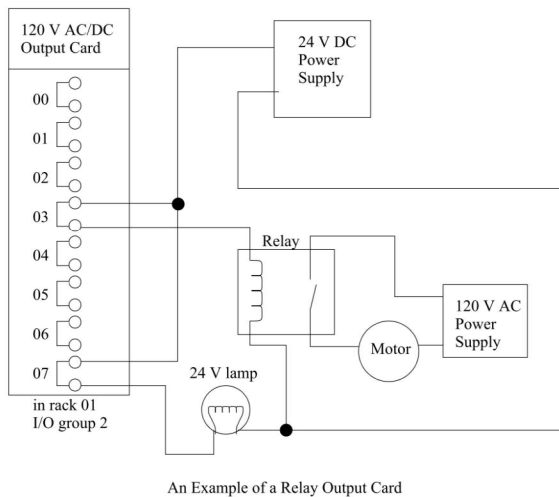
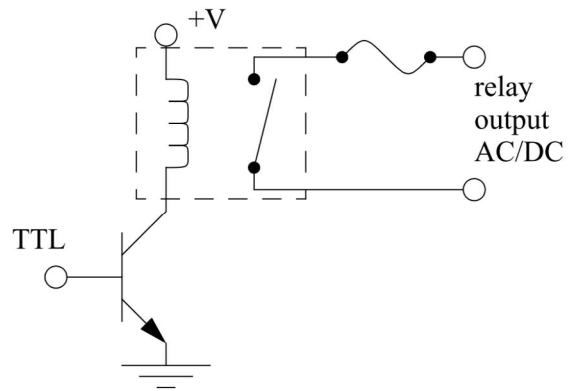
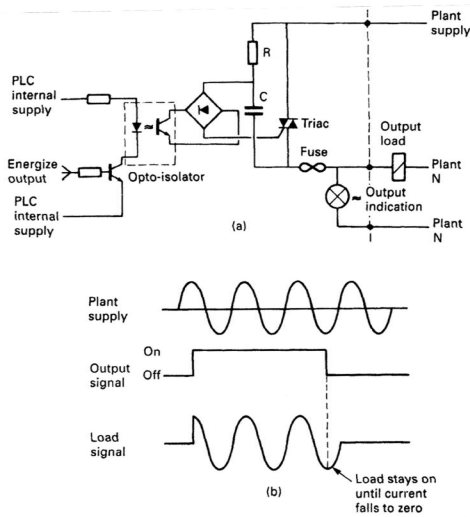
Figure 3.6 An Example of a 24Vdc Output Card With a Voltage Input (Sourcing)



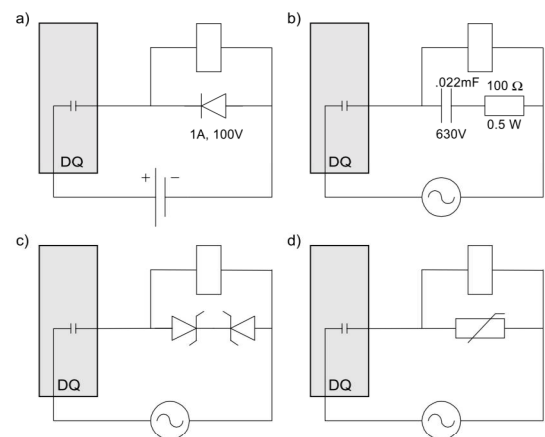
An Example of a 24Vdc Output Card (Sinking)

AC outputs

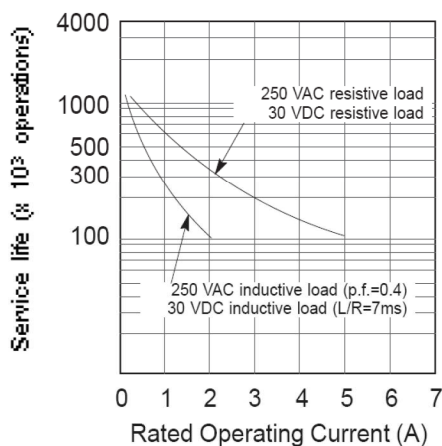




Protection of the output modules against overvoltage effects under inductive load



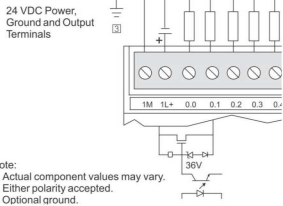
Lifetime of relay outputs



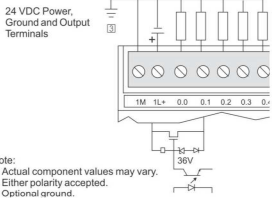
Parameters of output modules

- Rated voltage
- Size and number of groups
- Voltage and guaranteed time for galvanic isolation
- Switching on and off time
- Electrical characteristics - output current (per output/group/module)
- Current consumption (important for power supply selection)
- Operating conditions - temperature, humidity, vibrations

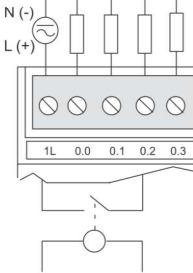
Output Features	
Number of integrated outputs	10 outputs
Output type	Solid state-MOSFET
Output Voltage	
Permissible range	20.4 to 28.8 VDC
Rated value	24 VDC
Logic 1 signal at maximum current	20 VDC, minimum
Logic 0 signal with 10 K Ω load	0.1 VDC, maximum
Output Current	
Logic 1 signal	0.75 A
Number of output groups	2
Number of outputs ON (maximum)	10
Per group - horizontal mounting (maximum)	5
Per group - vertical mounting (maximum)	5
Maximum current per common/group	3.75 A
Lamp load	5 W
ON state resistance (contact resistance)	0.3 Ω
Leakage current per point	10 μ A, maximum
Surge current	8 A for 100 ms, maximum
Overload protection	No



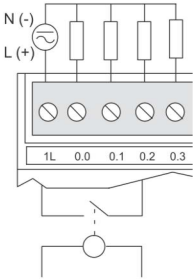
Isolation (Field Side to Logic)	
Optical isolation (galvanic)	500 VAC for 1 minute
Isolation resistance	-
Isolation coil to contact	-
Isolation between open contacts	-
In groups of	5 points
Inductive Load Clamping	
Repetitive	Energy dissipation < 0.5 LI ² x switching rate
Clamp voltage limits	L+ minus 48V
Output Delay	
Off to On (Q0.0 and Q0.1)	2 μ s, maximum
On to Off (Q0.0 and Q0.1)	10 μ s, maximum
Off to On (Q0.2 through Q1.1)	15 μ s, maximum
On to Off (Q0.2 through Q1.1)	100 μ s, maximum
Switching Frequency (Pulse Train Outputs)	
Q0.0 and I0.1	20 kHz, maximum
Cable Length	
Unshielded	150 m
Shielded	500 m



Output Features	
Number of integrated outputs	10 outputs
Output type	Relay, dry contact
Output Voltage	
Permissible range	5 to 30 VDC or 5 to 250 VAC
Rated value	-
Logic 1 signal at maximum current	-
Logic 0 signal with 10 K Ω load	-
Output Current	
Logic 1 signal	2.00 A
Number of output groups	3
Number of outputs ON (maximum)	10
Per group - horizontal mounting (maximum)	4/3/3
Per group - vertical mounting (maximum)	4/3/3
Maximum current per common/group	8 A
Lamp load	30 W DC/200 W AC
ON state resistance (contact resistance)	0.002 Ω , maximum when new
Leakage current per point	-
Surge current	7 A with contacts closed
Overload protection	No

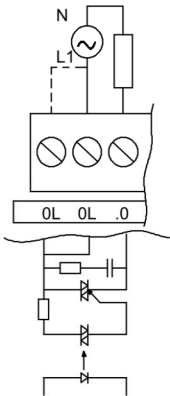


Isolation (Field Side to Logic)	
Optical isolation (galvanic)	-
Isolation resistance	100 M Ω , minimum when new
Isolation coil to contact	1500 VAC for 1 minute
Isolation between open contacts	750 VAC for 1 minute
In groups of	4 points/3 points/3 points
Inductive Load Clamping	
Repetitive	Energy dissipation < 0.5 LI ² x switching rate
Clamp voltage limits	-
Output Delay	
Off to On (Q0.0 and Q0.1)	-
On to Off (Q0.0 and Q0.1)	-
Off to On (Q0.2 through Q1.1)	-
On to Off (Q0.2 through Q1.1)	-
Switching Frequency (Pulse Train Outputs)	
Q0.0 and I0.1	1 Hz, maximum
Relay	
Switching delay	10 ms, maximum
Lifetime mechanical (no load)	10,000,000 open/close cycles
Lifetime contacts at rated load	100,000 open/close cycles
Cable Length	
Unshielded	150 m
Shielded	500 m



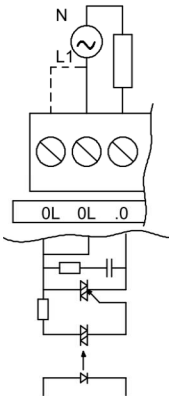
Type	Triac, zero-cross turn-on ²
Rated voltage	120/230 VAC
Voltage range	40 to 264 VAC (47 to 63 Hz)
24 VDC coil power voltage range	-
Surge current (max.)	5 A rms for 2 AC cycles
Logic 1 (min.)	L1 (~0.9 V rms)
Logic 0 (max.)	-
Rated current per point (max.)	0.5 A AC ³
Rated current per common (max.)	0.5 A AC
Leakage current (max.)	1.1 mA rms at 132 VAC and 1.8 mA rms at 264 VAC
Lamp load (max.)	60 W
Inductive clamp voltage	-
On state resistance (contact)	410 Ω max. when load current is less than 0.05 A

120/230 AC Output



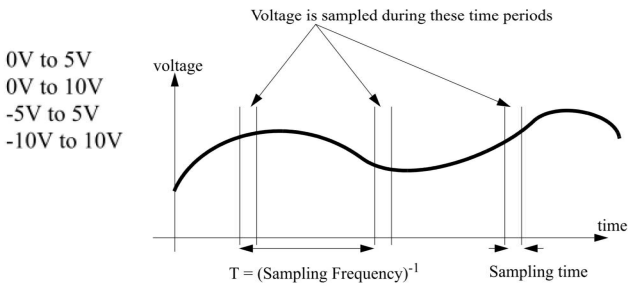
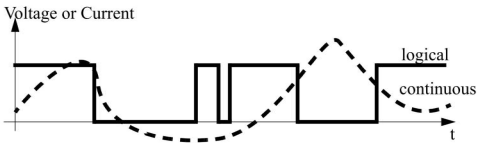
Isolation	
Optical (galvanic, field to logic)	1500 VAC for 1minute
Coil to logic	-
Coil to contact	-
Resistance (coil to contact)	-
Isolation groups	1 point
Delay Off to On/On to Off (max.)	0.2 ms + 1/2 AC cycle
Switching (max.)	-
Switching frequency (max.)	10 Hz
Lifetime mechanical cycles	-
Lifetime contacts	-
Output on simultaneously	All at 55° C (horizontal), All at 45° C (vertical)
Connecting two outputs in parallel	No
Cable length (max.)	
Shielded	500 m
Unshielded	150 m

120/230 AC Output

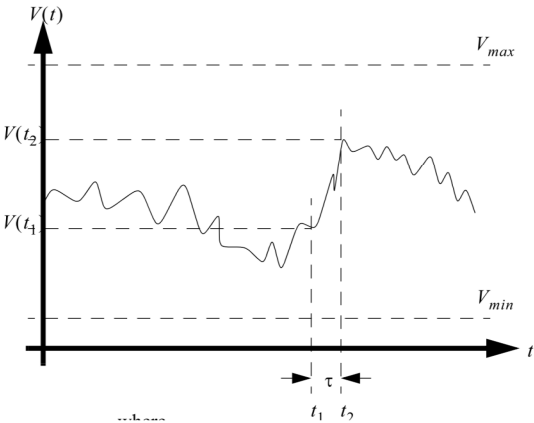


General	24 VDC Output		Relay Output		120/230 VAC Output
	0.75 A	5 A	2 A	10 A	
Type	Solid state-MOSFET ¹ (Sourcing)		Dry contact		Triac, zero-cross turn-on ²
Rated voltage	24 VDC		24 VDC or 250 VAC		120/230 VAC
Voltage range	20.4 to 28.8 VDC		5 to 30 VDC or 5 to 250 VAC	12 to 30 VDC or 12 to 250 VAC	40 to 264 VAC (47 to 63 Hz)
24 VDC coil power voltage range	-		20.4 to 28.8 VDC		-
Surge current (max.)	8 A for 100 ms	30 A	5 A for 4 s @ 10% duty cycle	15 A for 4 s @ 10% duty cycle	5 A rms for 2 AC cycles
Logic 1 (min.)	20 VDC		-		L1 (-0.9 V rms)
Logic 0 (max.)	0.1 VDC with 10 K Ω Load	0.2 VDC with 5 K Ω Load	-		-
Rated current per point (max.)	0.75 A	5 A	2.00 A	10 A resistive; 2 A DC inductive; 3 A AC inductive	0.5 A AC ³
Rated current per common (max.)	6 A	5 A	8 A	10 A	0.5 A AC
Leakage current (max.)	10 μ A	30 μ A	-		1.1 mA rms at 132 VAC and 1.8 mA rms at 264 VAC
Lamp load (max.)	5 W	50 W	30 W DC/ 200 W AC ^{4, 7}	100 W DC/ 1000 W AC	60 W

Continuous signal processing



Continuous signal processing



Continuous signal processing

Resolution
The digital output signal of the channel can only take some discrete values. So there is a quantization error that is identical in the whole range of the input signal. The maximum value of the quantization error is determined by the least significant *Least Significant Bit* (LSB) of the A/D converter register. The resolution is determined by this encoder quantisation error and is expressed as the weight of the least significant bit - e.g. 4 μ A/bit LSB.

No. of bits	Range	Error (%)
8	0-255	0.5
10	0-1023	0.1
12	0-4095	0.025

Resolution

$$R = 2^N = R_{max} - R_{min}$$

$$V_{ERROR} = \left(\frac{V_{max} - V_{min}}{2R} \right)$$

$$V_I = INT \left[\left(\frac{V_{in} - V_{min}}{V_{max} - V_{min}} \right) (R - 1) + R_{min} \right]$$

$$V_C = \left(\frac{V_I - R_{min}}{(R - 1)} \right) (V_{max} - V_{min}) + V_{min}$$

Accuracy

Depends on the tolerances of the components used as components for the module circuits. Accuracy is determined by determining the maximum difference between the expected and measured value, e.g:
 $\pm 0.25\%$ of the total scale for temperatures up to 25 $^{\circ}$ C,
 $\pm 0.5\%$ of total scale above 25 $^{\circ}$ C.

10 bit A/D converter 4.564V
-10V to 10V

Linearity

This is the difference between the measured changes for two adjacent channels by exactly one LSB bit - e.g. less than 1 LSB in the range from 4 to 20 mA.

Update Rate (Update Rate)

Conversion times

For a given number of input channels, e.g. 4 ms.

Common Mode Rejection Ratio

$$CMMR [dB] = 20 \cdot \lg \frac{U_c}{U_d}$$

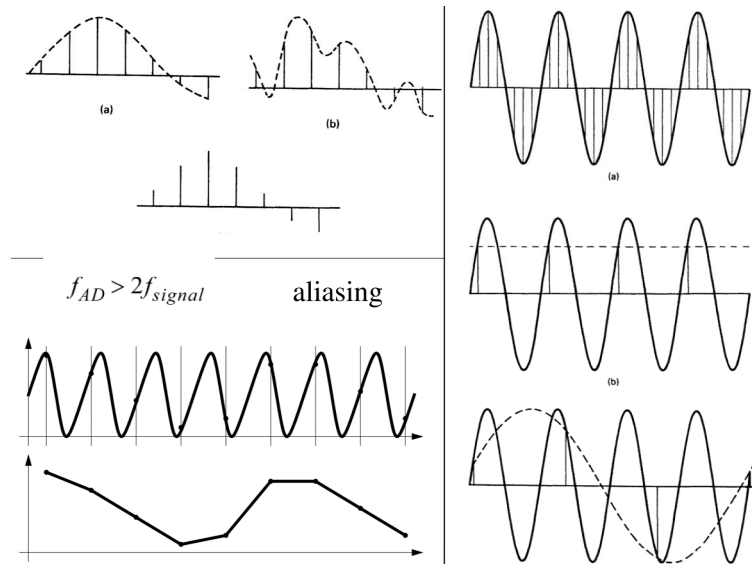
ratio of common voltage to differential voltage corresponding to its influence on the channel output. It defines the damping of common voltage in the channel - e.g. higher than 80 dB (10 000 :1) in the range from 0 to 1 kHz.

Cross-Channel Rejection Ratio

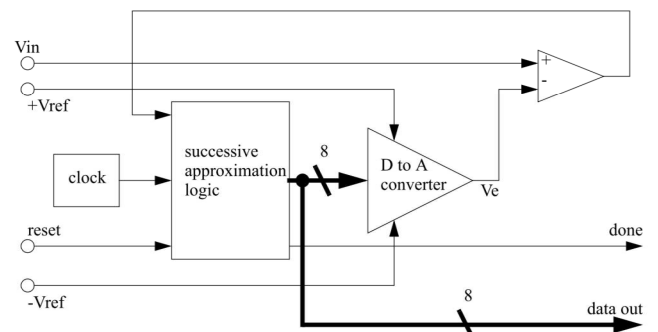
$$CCRR [dB] = 20 \cdot \lg \frac{U_x}{U_y}$$

Determines the influence of changes in the nearby channel U_x on the tested channel U_y .

Signal ratio in the nearby channel to the signal in the tested channel, e.g. greater than 100 dB (100,000 :1)

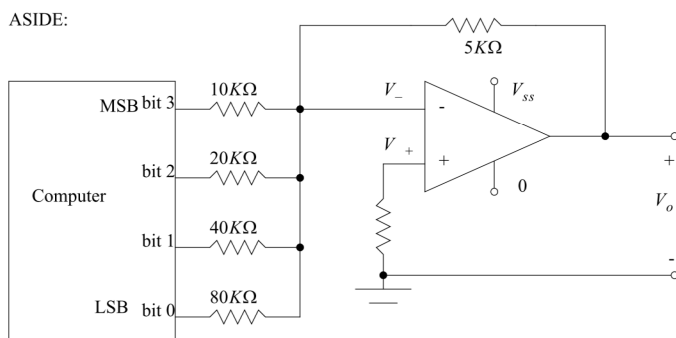


ADC with successive approximation

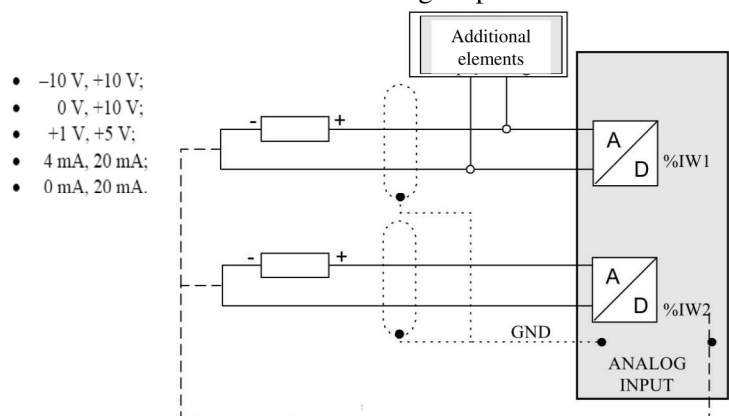


Binary-weighted DAC

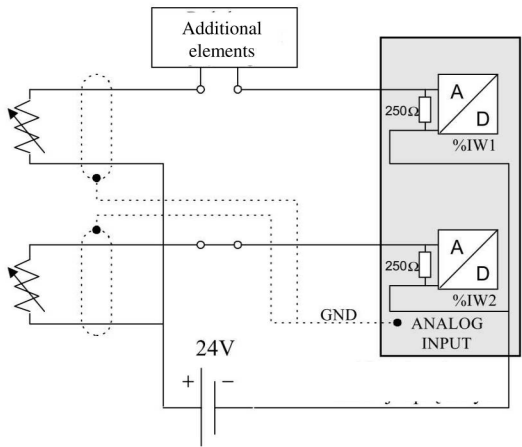
ASIDE:



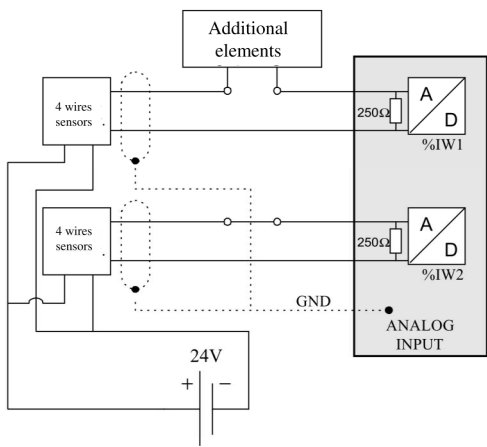
Connection diagram of potential converters with analogue module for voltage inputs



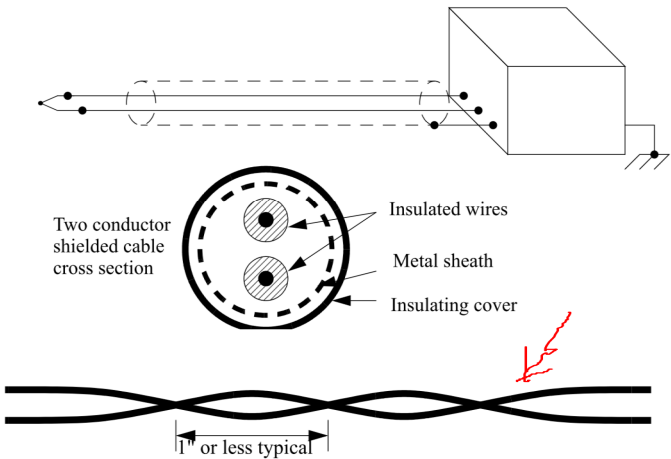
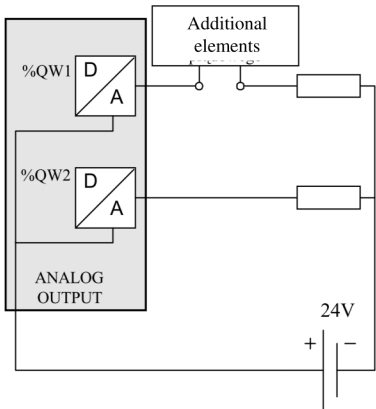
Connection diagram for 2-wire transmitters with single-ended current input module



Connection diagram of four-wire transmitters with differential current input module



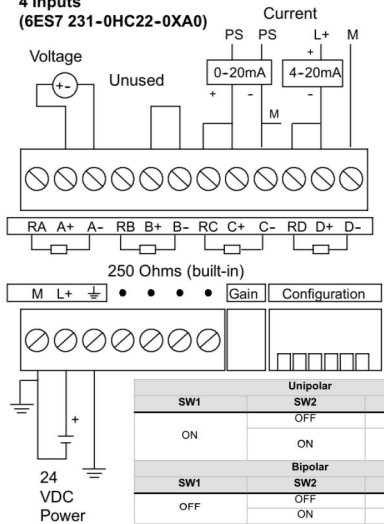
Load circuit diagram for analog output module for current signal:



Shielding of cables

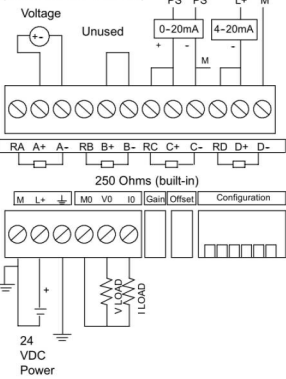
- **Electrostatic field** - a highly conductive shield, connected to GND or COM common point. Screens must not create closed circuits (feedback between noise source and input)
- **Electromagnetic field** - a screen with high magnetic permeability, does not need to be grounded. Also - use of twisted pair cable
- Most often - a screen used to eliminate both types of noise

EM 231 Analog Input, 4 Inputs (6ES7 231-0HC22-0XA0)

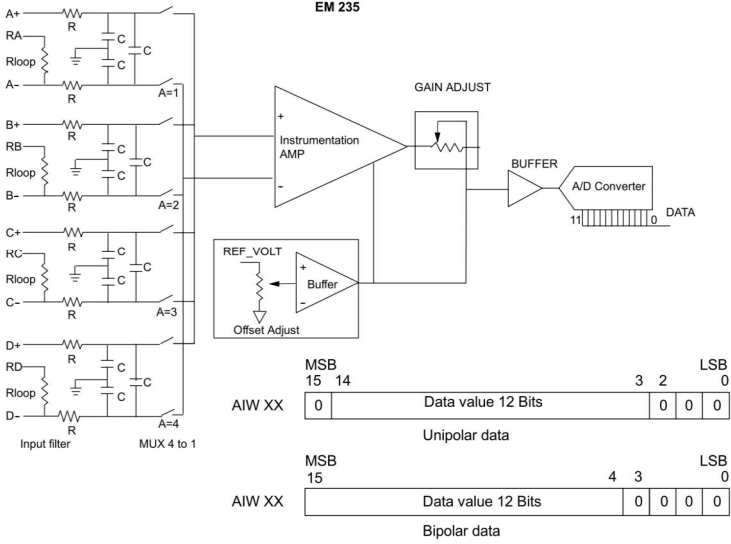


EM 235 Analog Combination

4 Inputs/1 Output
(6ES7 235-0KD22-0XA0)



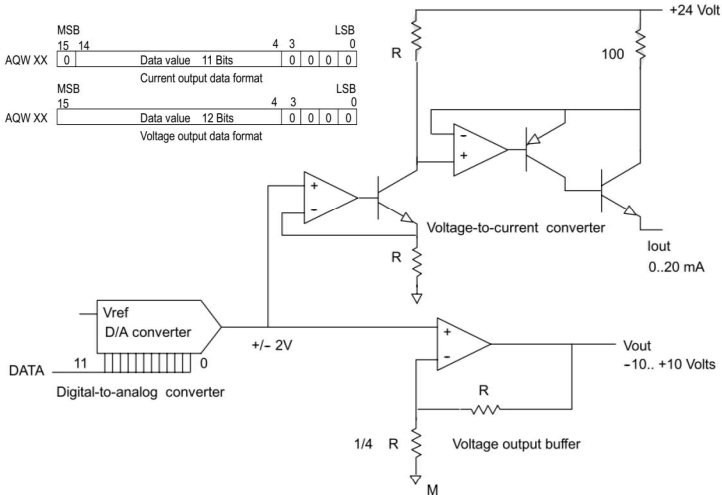
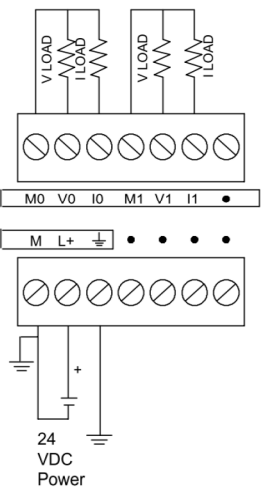
Unipolar						Full-Scale Input	Resolution
SW1	SW2	SW3	SW4	SW5	SW6		
ON	OFF	OFF	ON	OFF	ON	0 to 50 mV	12.5 μ V
OFF	ON	OFF	ON	OFF	ON	0 to 100 mV	25 μ V
ON	OFF	OFF	OFF	ON	ON	0 to 500 mV	125 μ V
OFF	ON	OFF	OFF	ON	ON	0 to 1 V	250 μ V
ON	OFF	OFF	OFF	OFF	ON	0 to 5 V	1.25 mV
ON	OFF	OFF	OFF	OFF	ON	0 to 20 mA	5 μ A
OFF	ON	OFF	OFF	OFF	ON	0 to 10 V	2.5 mV
Bipolar						Full-Scale Input	Resolution
SW1	SW2	SW3	SW4	SW5	SW6		
ON	OFF	OFF	ON	OFF	OFF	$\pm$ 25 mV	12.5 μ V
OFF	ON	OFF	ON	OFF	OFF	$\pm$ 50 mV	25 μ V
OFF	OFF	ON	ON	OFF	OFF	$\pm$ 100 mV	50 μ V
ON	OFF	OFF	OFF	ON	OFF	$\pm$ 250 mV	125 μ V
OFF	ON	OFF	OFF	ON	OFF	$\pm$ 500 mV	250 μ V
OFF	OFF	ON	OFF	ON	OFF	$\pm$ 1 V	500 μ V
ON	OFF	OFF	OFF	OFF	OFF	$\pm$ 2.5 V	1.25 mV
OFF	ON	OFF	OFF	OFF	OFF	$\pm$ 5 V	2.5 mV
OFF	OFF	ON	OFF	OFF	OFF	$\pm$ 10 V	5 mV



EM 231 Analog Input,
4 Inputs

Data word format	(See Figure A-14)
Bipolar, full-scale range	-32000 to +32000
Unipolar, full-scale range	0 to 32000
DC Input impedance	$\geq$ 10 M Ω voltage input 250 Ω current input
Input filter attenuation	-3 db at 3.1 KHz
Maximum input voltage	30 VDC
Maximum input current	32 mA
Resolution	Bipolar Unipolar
Isolation (field to logic)	None
Input type	Differential
Input ranges	Voltage Current
Input resolution	See Table A-20
Analog to digital conversion time	< 250 μ s
Analog input step response	1.5 ms to 95%
Common mode rejection	40 dB, DC to 60 Hz
Common mode voltage	Signal voltage plus common mode voltage must be $\leq \pm$ 12 V
24 VDC supply voltage range	20.4 to 28.8 VDC (Class 2, Limited Power, or sen

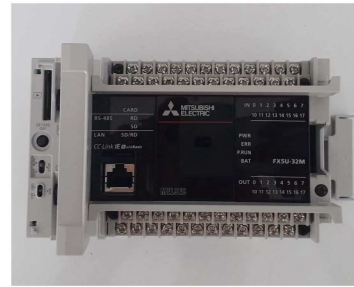
EM 232 Analog Output,
2 Outputs
(6ES7 232-0HB22-0XA0)



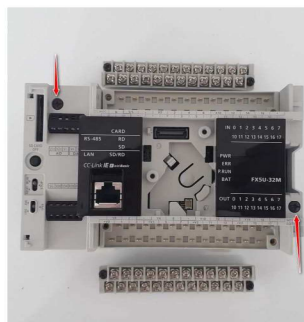
General	6ES7 232-0HB22-0XA0
Isolation (field to logic)	None
Signal range	
Voltage output	$\pm$ 10 V
Current output	0 to 20 mA
Resolution, full-scale	
Voltage	12 bits plus sign bit
Current	11 bits
Data word format	
Voltage	-32000 to +32000
Current	0 to +32000
Accuracy	
Worst case, 0° to 55° C	
Voltage output	$\pm$ 2% of full-scale
Current output	$\pm$ 2% of full-scale
Typical, 25° C	
Voltage output	$\pm$ 0.5% of full-scale
Current output	$\pm$ 0.5% of full-scale
Setting time	
Voltage output	100 μ s
Current output	2 mS
Maximum drive	
Voltage output	5000 Ω minimum
Current output	500 Ω maximum
24 VDC supply voltage range	20.4 to 28.8 VDC (Class 2, Limited Power, or sen



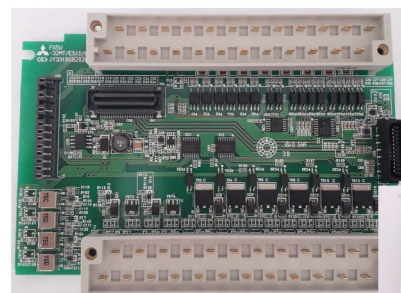
MELSEC iQ-F FX5U PLC produced by Mitsubishi Electric is a compact PLC that can be expanded with additional modules (e.g. positioning, temperature) on the right side of the controller and adapters (e.g. analog inputs/outputs, additional serial ports) on the left side of the controller.



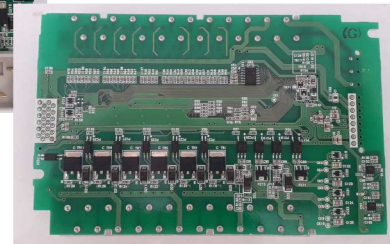
The left side shows the terminals for analog inputs/outputs, serial communication, SD card. On the left side there is a switch for controller operation mode (RUN/STOP/RESET) and a switch to select terminating resistor (110/330Ω) for the RS-485 port. On the right side there is a connector for modules connected by ribbon. In the middle there is a connector for additional driver expansion boards (RS-232/422/485 serial ports) and a connector for optional battery supporting RTC clock and Device Memory.



A big plus for maintenance staff is the ability to pull the PLC out of the control cabinet without having to undo the digital input and output terminals. Just unscrew 2 screws and pull the terminal towards yourself. View of the digital input/output terminal from the inside. We can see the metal plates which connect to the pins located on the controller board.



Digital input / output board ...



... and an SD card support board.



On the left analog I/O terminals, RS422/485 serial port terminal, RJ-45 socket for Ethernet port. The 32-bit R5F564MGDDFC microcontroller is used. On the right side you can see the capacitor that keeps the RTC clock powered. Above and below the capacitor are LEDs indicating the status of the digital input or output.

